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Multi-objective steady-state optimization for a complex bioprocess in glycerol metabolism

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ABSTRACT

This paper addresses the multi-objective steady-state optimization of a complex nonlinear system in glycerol metabolism. In the bioconversion system, we consider some mechanisms including the regulation for *dha* regulon, enzyme-catalytic kinetics, the inhibition of intracellular 3-hydroxypropionaldehyde (3-HPA), and so on. Firstly, we propose several new multi-objective steady-state optimization problems. These problems can simultaneously optimize any two of the following objectives: (1) minimization of the intracellular 3-HPA concentration; (2) maximization of the production rate of extracellular 1,3-propanediol (1,3-PDO); (3) maximization of the conversion rate of extracellular glycerol to extracellular 1,3-PDO; (4) maximization of the conversion rate of extracellular glycerol. Then a hybrid optimization method is proposed to solve the presented complex multi-objective problems. This optimization method couples a deterministic interior point approach and a stochastic genetic algorithm in the framework of normal-boundary intersection (NBI) method. After solving the proposed multi-objective problems of the addressed bioconversion system, we finally obtain their approximate Pareto optimal sets. The analysis for the attained optimization results is given. And a comparative study is also presented.

1. Introduction

1,3-propanediol (1,3-PDO) has widespread applications in many fields [1–4]. In the microbial production of 1,3-PDO, the synthesis of 1,3-PDO directly from glycerol metabolism has attracted an increasing interest worldwide since the 1980s [2,3,5–7]. In recent years, much research has been directed toward the improvement of glycerol metabolism process, including mathematical modeling, process analysis, parameter identification, process optimization, process control, etc. [2,3,5,6,8–36]. For example, Sun et al. [2] presented the advances in the 1,3-PDO production from glycerol. Laura et al. [3] addressed the evaluation for the ability of *Klebsiella pneumoniae* DSMZ 2026 to synthesize 1,3-PDO in batch processes. Lee et al. [5] reviewed and discussed the methods of transforming bioglycerol into product 1,3-PDO. To mathematically describe the kinetics of glycerol consume, product formation, and cell growth in a continuous bioconversion process, some mathematical descriptions have been presented in [8–10]. Veras et al. [12] carried out the conversion of pure and crude glycerol into 1,3-PDO using a UASB reactor. This work shows that the UASB reactor is highly effective and feasible for the 1,3-PDO production. Rodriguez et al. [13] presented a kinetic model to address the impact of initial concentration of raw glycerol on the product 1,3-PDO. Liu et al. [14] proposed a robust bi-objective strategy to optimally control the nonlinear system with uncertain time-delay and parameters in a microbial batch process of 1,3-PDO. Yuan et al. [15] used a globally parallel optimization algorithm to study the robust optimal control problem of a batch time-varying system in the 1,3-PDO production. Niu et al. [18] presented an optimal strategy to control a nonlinear switched system in microbial fed-batch

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process of 1,3-PDO. Narisetty et al. [19] enhanced 1,3-PDO production by using the methods of maintaining physical conditions and optimizing media compositions in the reactor. Gong et al. [20] proposed a distributionally robust method to identify the parameters of a class of time-delay system. This method was applied to a batch reactor of transforming glycerol into 1,3-PDO. Paranhos et al. [21] addressed the statistical optimization of 1,3-PDO and other products produced from crude glycerol by using an fluidized bed reactor. Yuan et al. [22] presented a robust identification method for nonlinear impulsive switched system in fed-batch culture process of glycerol producing 1,3-PDO by *Klebsiella pneumonia*. Xu and Li [23] established a three-layer programming formulation to determine a better common objective function for glycerol metabolism into 1,3-PDO by *Klebsiella pneumoniae*. This three-layer method has a more feasibility than the existing bi-level programming approach [23]. To deal with the difficulty of directly solving the parameter identification problem in the batch process of glycerol producing 1,3-PDO, Xu et al. [24] proposed a two-stage method to efficiently achieve this task. Xu et al. [25] addressed the multi-objective optimization of continuous bioconversion process of glycerol into 1,3-PDO. However, the used method in this study cannot obtain a better Pareto front of certain multi-objective problem (See the comparative study in Section 5). Therefore, a new solution algorithm should be proposed to improve the distribution of Pareto front. In addition, in the study of Xu et al. [25], the considered bioprocess is mathematically represented by a five-dimensional nonlinear system. And this representation involves only the kinetics of cell growth, extracellular glycerol, extracellular 1,3-PDO, extracellular acetic acid, and extracellular ethanol. However, in fact, the bioconversion process of glycerol producing 1,3-PDO is very complex and highly nonlinear system. It also includes the enzyme-catalytic on the reductive pathway, the regulation for dha regulon, the transport of substrate glycerol across cell membrane, the diffusion of product 1,3-PDO across cell membrane, and the inhibition of intracellular 3-hydroxypropionaldehyde (3-HPA). Additionally, intracellular 3-HPA is a highly toxic substance [37], its accumulation can cause cell growth to slow down or even stop, and has a certain inhibitory effect on the activities of key enzymes. Thus, it is necessary to minimize the concentration of intracellular 3-HPA during the complex bioconversion process.

In this study, to minimize the concentration of toxic intracellular 3-HPA, and can also maximize the production rate of extracellular 1,3-PDO, the conversion rate of extracellular substrate glycerol to extracellular 1,3-PDO, the conversion rate of extracellular substrate glycerol, we will deal with the multi-objective steady-state optimization problem of complex bioconversion process of glycerol metabolism producing 1,3-PDO.

In the following, we first represent the bioconversion process of glycerol metabolism into 1,3-PDO as a fourteen-dimensional nonlinear ODEs in Section 2. Then, the multi-objective steady-state optimization models of this biological process are proposed in Section 3. Section 4 proposes a hybrid optimization method within the framework of normal-boundary intersection (NBI) approach to solve the proposed multi-objective steady-state models. Section 5 presents the computation results obtained by the proposed hybrid optimization method and some discussions to them. Finally, some brief conclusions are given in Section 6.

2. Continuous bioconversion process of glycerol metabolism into 1,3-PDO

Based on previous works [6,37,38], the continuous bioconversion process of glycerol metabolism into 1,3-PDO by *Klebsiella pneumoniae* can be represented as the following fourteen-dimensional nonlinear ODEs:

$$\frac{\mathrm{d}X_{\mathrm{R}}}{\mathrm{d}t} = K_{X_{\mathrm{R}}} G_{\mathrm{D}} \frac{[\mathrm{O_t}]\left(X_{3-\mathrm{HPA}} + r_1\right)}{\left(X_{3-\mathrm{HPA}} + r_1\right) + r_2 X_{3-\mathrm{HPA}}} - \left(r_3 + \mu_{\mathrm{B}}\right) X_{\mathrm{R}} \tag{1}$$

$$\frac{\mathrm{d}X_{\mathrm{R1}}}{\mathrm{d}t} = K_{\mathrm{R}} X_{\mathrm{R}} - (r_4 + \mu_{\mathrm{B}}) X_{\mathrm{R1}} - r_5[(3-\mathrm{HPA})_{\mathrm{n}} R] \tag{2}$$

$$\frac{\mathrm{d}X_{\mathrm{GDHt}}}{\mathrm{d}t} = K_{\mathrm{m}}^{\mathrm{GDHt}} G_{\mathrm{D}} \frac{[\mathrm{O_t}]\left(X_{3-\mathrm{HPA}} + r_1\right)}{\left(X_{3-\mathrm{HPA}} + r_1\right) + r_2 X_{3-\mathrm{HPA}}} - \left(r_7 + \mu_{\mathrm{B}}\right) X_{\mathrm{GDHt}} \tag{3}$$

$$\frac{\mathrm{d}X_{\mathrm{PDOR}}}{\mathrm{d}t} = K_{\mathrm{m}}^{\mathrm{PDOR}} G_{\mathrm{D}} \frac{[\mathrm{O_t}]\left(X_{3-\mathrm{HPA}} + r_1\right)}{\left(X_{3-\mathrm{HPA}} + r_1\right) + r_2 X_{3-\mathrm{HPA}}} - \left(r_8 + \mu_{\mathrm{B}}\right) X_{\mathrm{PDOR}} \tag{4}$$

$$\frac{\mathrm{d}X_{[\mathrm{GDHt}]}}{\mathrm{d}t} = K_{\mathrm{GDHt}} X_{\mathrm{GDHt}} - \left(r_9 + \mu_{\mathrm{B}}\right) X_{[\mathrm{GDHt}]} \tag{5}$$

$$\frac{\mathrm{d}X_{[\mathrm{PDOR}]}}{\mathrm{d}t} = K_{\mathrm{PDOR}} X_{\mathrm{PDOR}} - \left(r_{10} + \mu_{\mathrm{B}}\right) X_{[\mathrm{PDOR}]} \tag{6}$$

$$\frac{\mathrm{d}X_{\mathrm{B}}}{\mathrm{d}t} = (\mu_{\mathrm{B}} - D) X_{\mathrm{B}} \tag{7}$$

$$\frac{\mathrm{d}X_{\mathrm{Se}}}{\mathrm{d}t} = D\left(X_{\mathrm{SF}} - X_{\mathrm{Se}}\right) - q_{\mathrm{S}} X_{\mathrm{B}} \tag{8}$$

$$\frac{\mathrm{d}X_{\mathrm{Si}}}{\mathrm{d}t} = \frac{1}{r_{11}}\left[\frac{r_{12} X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{13}} + \frac{1}{r_{14}}\left(X_{\mathrm{Se}} - X_{\mathrm{Si}}\right) - q_{\mathrm{S}}\right] - \mu_{\mathrm{B}} X_{\mathrm{Si}} \tag{9}$$

$$\begin{aligned}\frac{\mathrm{d}X_{3-\mathrm{HPA}}}{\mathrm{d}t} &= r_{15} X_{[\mathrm{GDHt}]} \frac{X_{\mathrm{Si}}}{r_{17}\left(1 + \frac{X_{3-\mathrm{HPA}}}{r_{19}}\right) + X_{\mathrm{Si}}} - r_{16} X_{[\mathrm{PDOR}]} \frac{X_{3-\mathrm{HPA}}}{r_{18} + X_{3-\mathrm{HPA}}\left(1 + \frac{X_{3-\mathrm{HPA}}}{r_{20}}\right)} \\ &\quad - \mu_{\mathrm{B}} X_{3-\mathrm{HPA}} - r_{21} r_5 \frac{r_6 X_{3-\mathrm{HPA}}}{X_{3-\mathrm{HPA}} + r_1}\end{aligned} \tag{10}$$

$$\frac{\mathrm{d}X_{\mathrm{PDi}}}{\mathrm{d}t} = r_{16} X_{[\mathrm{PDOR}]} \frac{X_{3-\mathrm{HPA}}}{r_{18} + X_{3-\mathrm{HPA}}\left(1 + \frac{X_{3-\mathrm{HPA}}}{r_{20}}\right)} - r_{22}\left(X_{\mathrm{PDi}} - X_{\mathrm{PDe}}\right) - \mu_{\mathrm{B}} X_{\mathrm{PDi}} \tag{11}$$

$$\frac{\mathrm{d}X_{\mathrm{PDe}}}{\mathrm{d}t} = q_{\mathrm{PDe}}X_{\mathrm{B}} - DX_{\mathrm{PDe}} \tag{12}$$

$$\frac{\mathrm{d}X_{\mathrm{HAc}}}{\mathrm{d}t} = q_{\mathrm{HAc}}X_{\mathrm{B}} - DX_{\mathrm{HAc}} \tag{13}$$

$$\frac{\mathrm{d}X_{\mathrm{EtOH}}}{\mathrm{d}t} = q_{\mathrm{EtOH}}X_{\mathrm{B}} - DX_{\mathrm{EtOH}} \tag{14}$$

$$[(3-\mathrm{HPA})_{\mathrm{n}}R] = \frac{r_6 X_{3-\mathrm{HPA}}}{X_{3-\mathrm{HPA}} + r_1} \tag{15}$$

$$\mu_{\mathrm{B}} = r_{23}\frac{X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{24}}\left(1 - \frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1 - \frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1 - \frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1 - \frac{X_{\mathrm{EtOH}}}{r_{28}}\right) \tag{16}$$

$$q_{\mathrm{S}} = r_{29} + \frac{\mu_{\mathrm{B}}}{r_{30}} + r_{31}\frac{X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{32}} \tag{17}$$

$$q_{\mathrm{PDe}} = r_{33} + r_{34}\mu_{\mathrm{B}} + r_{35}\frac{X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{36}} \tag{18}$$

$$q_{\mathrm{HAc}} = r_{37} + r_{38}\mu_{\mathrm{B}} + r_{39}\frac{X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{40}} \tag{19}$$

$$q_{\mathrm{EtOH}} = q_{\mathrm{S}}\left(\frac{r_{41}}{r_{43} + DX_{\mathrm{Se}}} + \frac{r_{42}}{r_{44} + DX_{\mathrm{Se}}}\right) \tag{20}$$

where t is the bioconversion time (h); X_{R} and X_{R1} denote the concentrations of mRNA coding repressor and free repressor, respectively (mmol/L); $[(3-\mathrm{HPA})_{\mathrm{n}}R]$ denotes the holorepressor concentration; G_{D} is the gene dosage; $[\mathrm{O_t}]$ is the total operator concentration (mmol/L); X_{GDHt} and X_{PDOR} denote the mRNA concentrations of coding enzymes glycerol dehydratase and 1,3-propanediol oxidoreductase, respectively (mmol/L); $X_{[\mathrm{GDHt}]}$ and $X_{[\mathrm{PDOR}]}$ are the concentrations of enzymes glycerol dehydratase and 1,3-propanediol oxidoreductase, respectively (mmol/L); X_{B} represents the biomass (g/L); D denotes the dilution rate (h^{-1}); X_{SF} is the substrate (glycerol) concentration in feed (mmol/L); X_{Se} and X_{Si} denote the concentrations of extracellular and intracellular glycerol in reactor, respectively (mmol/L); $X_{3-\mathrm{HPA}}$ denotes the concentration of intracellular 3-HPA (mmol/L); X_{PDi} and X_{PDe} are the concentrations of intracellular and extracellular 1,3-PDO, respectively (mmol/L); X_{HAc} is the concentration of extracellular acetic acid (mmol/L); X_{EtOH} represents the concentration of extracellular ethanol (mmol/L); μ_{B} represents the specific growth rate of cells (h^{-1}); q_{S} is the specific consumption rate of substrate glycerol (mmol/(g·h)); q_{PDe}, q_{HAc}, q_{EtOH} are the specific formation rates of extracellular 1,3-PDO, extracellular acetic acid, and extracellular ethanol, respectively (mmol/(g·h)); and the definitions of the remain parameters shown in Eqs. (1)–(20) can be found in [6,37,38], and their values are listed in Appendix.

By substituting Eqs. (15)–(20) into Eqs. (1)–(14) and introducing the following transformations:

$$m_{\mathrm{R}} = \frac{X_{\mathrm{R}}}{K_{M_{\mathrm{R}}}G_{\mathrm{D}}[\mathrm{O_t}]}$$

$$r_{45} = K_{\mathrm{R}}K_{M_{\mathrm{R}}}G_{\mathrm{D}}[\mathrm{O_t}]$$

$$m_{\mathrm{GDHt}} = \frac{X_{\mathrm{GDHt}}}{K_{\mathrm{m}}^{\mathrm{GDHt}}G_{\mathrm{D}}[\mathrm{O_t}]}$$

$$r_{46} = K_{\mathrm{m}}^{\mathrm{GDHt}}K_{\mathrm{GDHt}}G_{\mathrm{D}}[\mathrm{O_t}]$$

$$m_{\mathrm{PDOR}} = \frac{X_{\mathrm{PDOR}}}{K_{\mathrm{m}}^{\mathrm{PDOR}}G_{\mathrm{D}}[\mathrm{O_t}]}$$

$$r_{47} = K_{\mathrm{m}}^{\mathrm{PDOR}}K_{\mathrm{PDOR}}G_{\mathrm{D}}[\mathrm{O_t}]$$

we can rewrite Eqs. (1)–(14) as:

$$\frac{\mathrm{d}m_{\mathrm{R}}}{\mathrm{d}t} = \frac{X_{3-\mathrm{HPA}} + r_1}{X_{3-\mathrm{HPA}} + r_1 + r_2X_{3-\mathrm{HPA}}} - m_{\mathrm{R}}\left[r_3 + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{24}}\left(1 - \frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1 - \frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1 - \frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1 - \frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] \tag{21}$$

$$\frac{\mathrm{d}X_{\mathrm{R1}}}{\mathrm{d}t} = r_{45}m_{\mathrm{R}} - X_{\mathrm{R1}}\left[r_4 + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{24}}\left(1 - \frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1 - \frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1 - \frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1 - \frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] - \frac{r_5r_6X_{3-\mathrm{HPA}}}{X_{3-\mathrm{HPA}} + r_1} \tag{22}$$

$$\frac{\mathrm{d}m_{\mathrm{GDHt}}}{\mathrm{d}t} = \frac{X_{3-\mathrm{HPA}} + r_1}{X_{3-\mathrm{HPA}} + r_1 + r_2X_{3-\mathrm{HPA}}} - m_{\mathrm{GDHt}}\left[r_7 + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{24}}\left(1 - \frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1 - \frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1 - \frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1 - \frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] \tag{23}$$

$$\frac{\mathrm{d}m_{\mathrm{PDOR}}}{\mathrm{d}t} = \frac{X_{3-\mathrm{HPA}} + r_1}{X_{3-\mathrm{HPA}} + r_1 + r_2X_{3-\mathrm{HPA}}} - m_{\mathrm{PDOR}}\left[r_8 + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{24}}\left(1 - \frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1 - \frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1 - \frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1 - \frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] \tag{24}$$

$$\frac{\mathrm{d}X_{[\mathrm{GDHt}]}}{\mathrm{d}t} = r_{46}m_{\mathrm{GDHt}} - X_{[\mathrm{GDHt}]}\left[r_9 + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{24}}\left(1 - \frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1 - \frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1 - \frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1 - \frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] \tag{25}$$

$$\frac{\mathrm{d}X_{[\mathrm{PDOR}]}}{\mathrm{d}t} = r_{47}m_{\mathrm{PDOR}} - X_{[\mathrm{PDOR}]}\left[r_{10} + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{24}}\left(1 - \frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1 - \frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1 - \frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1 - \frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] \tag{26}$$

$$\frac{\mathrm{d}X_{\mathrm{B}}}{\mathrm{d}t} = \left[\frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{24}}\left(1 - \frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1 - \frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1 - \frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1 - \frac{X_{\mathrm{EtOH}}}{r_{28}}\right) - D\right]X_{\mathrm{B}} \tag{27}$$

$$\frac{\mathrm{d}X_{\mathrm{Se}}}{\mathrm{d}t} = D\left(X_{\mathrm{SF}} - X_{\mathrm{Se}}\right) - \left[r_{29} + \frac{r_{23}X_{\mathrm{Se}}}{r_{30}\left(X_{\mathrm{Se}} + r_{24}\right)}\left(1 - \frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1 - \frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1 - \frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1 - \frac{X_{\mathrm{EtOH}}}{r_{28}}\right) + \frac{r_{31}X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{32}}\right]X_{\mathrm{B}} \tag{28}$$

$$\frac{\mathrm{d}X_{\mathrm{Si}}}{\mathrm{d}t} = \frac{1}{r_{11}}\left\{\frac{r_{12}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{13}} + \frac{1}{r_{14}}\left(X_{\mathrm{Se}}-X_{\mathrm{Si}}\right) - \left[r_{29} + \frac{r_{23}X_{\mathrm{Se}}}{r_{30}\left(X_{\mathrm{Se}}+r_{24}\right)}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right.\right.$$
$$\left.\left.+\frac{r_{31}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{32}}\right]\right\} - \frac{r_{23}X_{\mathrm{Si}}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right) \tag{29}$$

$$\frac{\mathrm{d}X_{3-\mathrm{HPA}}}{\mathrm{d}t} = \frac{r_{15}X_{[\mathrm{GDHt}]}X_{\mathrm{Si}}}{r_{17}\left(1+\frac{X_{3-\mathrm{HPA}}}{r_{19}}\right)+X_{\mathrm{Si}}} - \frac{r_{16}X_{[\mathrm{PDOR}]}X_{3-\mathrm{HPA}}}{r_{18}+X_{3-\mathrm{HPA}}\left(1+\frac{X_{3-\mathrm{HPA}}}{r_{20}}\right)} - \frac{r_{23}X_{3-\mathrm{HPA}}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)$$
$$\times\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right) - \frac{r_5r_6r_{21}X_{3-\mathrm{HPA}}}{X_{3-\mathrm{HPA}}+r_1} \tag{30}$$

$$\frac{\mathrm{d}X_{\mathrm{PDi}}}{\mathrm{d}t} = \frac{r_{16}X_{[\mathrm{PDOR}]}X_{3-\mathrm{HPA}}}{r_{18}+X_{3-\mathrm{HPA}}\left(1+\frac{X_{3-\mathrm{HPA}}}{r_{20}}\right)} - r_{22}\left(X_{\mathrm{PDi}}-X_{\mathrm{PDe}}\right) - \frac{r_{23}X_{\mathrm{PDi}}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right) \tag{31}$$

$$\frac{\mathrm{d}X_{\mathrm{PDe}}}{\mathrm{d}t} = \left[r_{33} + \frac{r_{23}r_{34}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right) + r_{35}\frac{X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{36}}\right]X_{\mathrm{B}} - DX_{\mathrm{PDe}} \tag{32}$$

$$\frac{\mathrm{d}X_{\mathrm{HAc}}}{\mathrm{d}t} = \left[r_{37} + \frac{r_{23}r_{38}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right) + r_{39}\frac{X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{40}}\right]X_{\mathrm{B}} - DX_{\mathrm{HAc}} \tag{33}$$

$$\frac{\mathrm{d}X_{\mathrm{EtOH}}}{\mathrm{d}t} = \left[r_{29} + \frac{r_{23}X_{\mathrm{Se}}}{r_{30}\left(X_{\mathrm{Se}}+r_{24}\right)}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right) + \frac{r_{31}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{32}}\right]$$
$$\times\left(\frac{r_{41}}{r_{43}+DX_{\mathrm{Se}}} + \frac{r_{42}}{r_{44}+DX_{\mathrm{Se}}}\right)X_{\mathrm{B}} - DX_{\mathrm{EtOH}} \tag{34}$$

In Eqs. (22), (25) and (26), the values of r_{45}, r_{46}, and r_{47} are listed in Appendix.

3. Multi-objective steady-state optimization models of glycerol metabolism

In this section, we propose several multi-objective steady-state optimization models for continuous bioconversion process of glycerol metabolism into 1,3-PDO. These models can simultaneously optimize any two of the following objectives: (1) minimization of the intracellular 3-HPA concentration $X_{3-\mathrm{HPA}}$; (2) maximization of the production rate DX_{PDe} of extracellular 1,3-PDO; (3) maximization of the conversion rate $X_{\mathrm{PDe}}/(X_{\mathrm{SF}}-X_{\mathrm{Se}})$ of extracellular glycerol into extracellular 1,3-PDO; (4) maximization of the conversion rate $(X_{\mathrm{SF}}-X_{\mathrm{Se}})/X_{\mathrm{SF}}$ of extracellular glycerol.

Considering the aforementioned four optimization indexes and assuming the studied bioconversion process operating at steady-state, the following six multi-objective steady-state optimization models are proposed.

Multi-objective model A:

$$\min \quad J_{11} = X_{3-\mathrm{HPA}}$$
$$\max \quad J_{12} = DX_{\mathrm{PDe}}$$

subject to satisfying:

$$\frac{X_{3-\mathrm{HPA}}+r_1}{X_{3-\mathrm{HPA}}+r_1+r_2X_{3-\mathrm{HPA}}} - m_{\mathrm{R}}\left[r_3 + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] = 0 \tag{35}$$

$$r_{45}m_{\mathrm{R}} - X_{\mathrm{R1}}\left[r_4 + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] - \frac{r_5r_6X_{3-\mathrm{HPA}}}{X_{3-\mathrm{HPA}}+r_1} = 0 \tag{36}$$

$$\frac{X_{3-\mathrm{HPA}}+r_1}{X_{3-\mathrm{HPA}}+r_1+r_2X_{3-\mathrm{HPA}}} - m_{\mathrm{GDHt}}\left[r_7 + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] = 0 \tag{37}$$

$$\frac{X_{3-\mathrm{HPA}}+r_1}{X_{3-\mathrm{HPA}}+r_1+r_2X_{3-\mathrm{HPA}}} - m_{\mathrm{PDOR}}\left[r_8 + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] = 0 \tag{38}$$

$$r_{46}m_{\mathrm{GDHt}} - X_{[\mathrm{GDHt}]}\left[r_9 + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] = 0 \tag{39}$$

$$r_{47}m_{\mathrm{PDOR}} - X_{[\mathrm{PDOR}]}\left[r_{10} + \frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right] = 0 \tag{40}$$

$$\left[\frac{r_{23}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right) - D\right]X_{\mathrm{B}} = 0 \tag{41}$$

$$D\left(X_{\mathrm{SF}}-X_{\mathrm{Se}}\right) - \left[r_{29} + \frac{r_{23}X_{\mathrm{Se}}}{r_{30}\left(X_{\mathrm{Se}}+r_{24}\right)}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right) + \frac{r_{31}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{32}}\right]X_{\mathrm{B}} = 0 \tag{42}$$

$$\frac{1}{r_{11}}\left\{\frac{r_{12}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{13}} + \frac{1}{r_{14}}\left(X_{\mathrm{Se}}-X_{\mathrm{Si}}\right) - \left[r_{29} + \frac{r_{23}X_{\mathrm{Se}}}{r_{30}\left(X_{\mathrm{Se}}+r_{24}\right)}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)\right.\right.$$

$$+\frac{r_{31}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{32}}\Bigg]\Bigg\}-\frac{r_{23}X_{\mathrm{Si}}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)=0 \tag{43}$$

$$\frac{r_{15}X_{[\mathrm{GDHt}]}X_{\mathrm{Si}}}{r_{17}\left(1+\frac{X_{3-\mathrm{HPA}}}{r_{19}}\right)+X_{\mathrm{Si}}}-\frac{r_{16}X_{[\mathrm{PDOR}]}X_{3-\mathrm{HPA}}}{r_{18}+X_{3-\mathrm{HPA}}\left(1+\frac{X_{3-\mathrm{HPA}}}{r_{20}}\right)}-\frac{r_{23}X_{3-\mathrm{HPA}}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)$$
$$\times\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)-\frac{r_5r_6r_{21}X_{3-\mathrm{HPA}}}{X_{3-\mathrm{HPA}}+r_1}=0 \tag{44}$$

$$\frac{r_{16}X_{[\mathrm{PDOR}]}X_{3-\mathrm{HPA}}}{r_{18}+X_{3-\mathrm{HPA}}\left(1+\frac{X_{3-\mathrm{HPA}}}{r_{20}}\right)}-r_{22}\left(X_{\mathrm{PDi}}-X_{\mathrm{PDe}}\right)-\frac{r_{23}X_{\mathrm{PDi}}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)=0 \tag{45}$$

$$\left[r_{33}+\frac{r_{23}r_{34}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)+r_{35}\frac{X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{36}}\right]X_{\mathrm{B}}-DX_{\mathrm{PDe}}=0 \tag{46}$$

$$\left[r_{37}+\frac{r_{23}r_{38}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{24}}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)+r_{39}\frac{X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{40}}\right]X_{\mathrm{B}}-DX_{\mathrm{HAc}}=0 \tag{47}$$

$$\left[r_{29}+\frac{r_{23}X_{\mathrm{Se}}}{r_{30}\left(X_{\mathrm{Se}}+r_{24}\right)}\left(1-\frac{X_{\mathrm{Se}}}{r_{25}}\right)\left(1-\frac{X_{\mathrm{PDe}}}{r_{26}}\right)\left(1-\frac{X_{\mathrm{HAc}}}{r_{27}}\right)\left(1-\frac{X_{\mathrm{EtOH}}}{r_{28}}\right)+\frac{r_{31}X_{\mathrm{Se}}}{X_{\mathrm{Se}}+r_{32}}\right]$$
$$\times\left(\frac{r_{41}}{r_{43}+DX_{\mathrm{Se}}}+\frac{r_{42}}{r_{44}+DX_{\mathrm{Se}}}\right)X_{\mathrm{B}}-DX_{\mathrm{EtOH}}=0 \tag{48}$$

$$0.05\le D\le 0.5 \tag{49}$$

$$0\le X_{\mathrm{SF}}\le 2039 \tag{50}$$

$$0\le m_{\mathrm{R}}\le 10 \tag{51}$$

$$0\le X_{\mathrm{R1}}\le 10 \tag{52}$$

$$0\le m_{\mathrm{GDHt}}\le 10 \tag{53}$$

$$0\le m_{\mathrm{PDOR}}\le 10 \tag{54}$$

$$0\le X_{[\mathrm{GDHt}]}\le 10 \tag{55}$$

$$0\le X_{[\mathrm{PDOR}]}\le 10 \tag{56}$$

$$0.05\le X_{\mathrm{B}}\le 5 \tag{57}$$

$$0\le X_{\mathrm{Se}}\le 2039 \tag{58}$$

$$0\le X_{\mathrm{Si}}\le 2039 \tag{59}$$

$$0\le X_{3-\mathrm{HPA}}\le 1000 \tag{60}$$

$$0\le X_{\mathrm{PDi}}\le 939.5 \tag{61}$$

$$0\le X_{\mathrm{PDe}}\le 939.5 \tag{62}$$

$$0\le X_{\mathrm{HAc}}\le 1026 \tag{63}$$

$$0\le X_{\mathrm{EtOH}}\le 360.9 \tag{64}$$

Multi-objective model B:

$$\min \quad J_{21}=X_{3-\mathrm{HPA}}$$
$$\max \quad J_{22}=\frac{X_{\mathrm{PDe}}}{X_{\mathrm{SF}}-X_{\mathrm{Se}}}$$

subject to satisfying: Eqs. (35)–(64).

Multi-objective model C:

$$\min \quad J_{31}=X_{3-\mathrm{HPA}}$$
$$\max \quad J_{32}=\frac{X_{\mathrm{SF}}-X_{\mathrm{Se}}}{X_{\mathrm{SF}}}$$

subject to satisfying: Eqs. (35)–(64).

Multi-objective model D:

$$\max \quad J_{41}=DX_{\mathrm{PDe}}$$
$$\max \quad J_{42}=\frac{X_{\mathrm{PDe}}}{X_{\mathrm{SF}}-X_{\mathrm{Se}}}$$

subject to satisfying: Eqs. (35)–(64).

Multi-objective model E:

$$\max \quad J_{51} = DX_{\mathrm{PDe}}$$
$$\max \quad J_{52} = \frac{X_{\mathrm{SF}} - X_{\mathrm{Se}}}{X_{\mathrm{SF}}}$$

subject to satisfying: Eqs. (35)–(64).

Multi-objective model F:

$$\max \quad J_{61} = \frac{X_{\mathrm{PDe}}}{X_{\mathrm{SF}} - X_{\mathrm{Se}}}$$
$$\max \quad J_{62} = \frac{X_{\mathrm{SF}} - X_{\mathrm{Se}}}{X_{\mathrm{SF}}}$$

subject to satisfying: Eqs. (35)–(64).

We can see that each of multi-objective models A–F is a multi-objective nonlinear programming with complex equality constraints and bounded inequality constraints. In these models, Eqs. (35)–(48) denote the steady-state conditions, while Eqs. (49)–(64) denote the bounds for the variables D, X_{SF}, m_{R}, X_{R1}, m_{GDHt}, m_{PDOR}, $X_{[\mathrm{GDHt}]}$, $X_{[\mathrm{PDOR}]}$, X_{B}, X_{Se}, X_{Si}, $X_{3-\mathrm{HPA}}$, X_{PDi}, X_{PDe}, X_{HAc}, and X_{EtOH}, respectively. Eqs. (35)–(64) determine the feasible space of multi-objective models A–F, while the set of all possible values of the two objective functions in every multi-objective model constitutes the corresponding objective space.

Note that no feasible solutions can simultaneously optimize the two objectives in each of multi-objective models A–F. Therefore, the so-called Pareto optimal solutions are needed to be found in the study of a multi-objective problem. For the proposed multi-objective models A–F, the Pareto optimal solutions are those feasible solutions that cannot be improved in any of the two objective functions without degrading the other objective function. The set of the corresponding objective values to Pareto optimal solutions is called the Pareto front.

4. Hybrid optimization method for the proposed multi-objective optimization models

In this section, we will propose a hybrid optimization method in the framework of the NBI approach to solve the multi-objective models A–F proposed in Section 3.

For convenience, the proposed multi-objective models A–F are first rewritten as:

$$\max \quad F_{i1}(\boldsymbol{y}) \tag{65}$$
$$\max \quad F_{i2}(\boldsymbol{y}) \tag{66}$$

subject to satisfying:

$$H_j(\boldsymbol{y}) = 0, j = 1, 2, \ldots, 14 \tag{67}$$
$$G_k(\boldsymbol{y}) \le 0, k = 1, 2, \ldots, 32 \tag{68}$$

where

$$i = 1, 2, \ldots, 6$$
$$\boldsymbol{y} = (y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9, y_{10}, y_{11}, y_{12}, y_{13}, y_{14}, y_{15}, y_{16})^{\mathrm{T}}$$
$$= (m_{\mathrm{R}}, X_{\mathrm{R1}}, m_{\mathrm{GDHt}}, m_{\mathrm{PDOR}}, X_{[\mathrm{GDHt}]}, X_{[\mathrm{PDOR}]}, X_{\mathrm{B}}, X_{\mathrm{Se}}, X_{\mathrm{Si}}, X_{3-\mathrm{HPA}}, X_{\mathrm{PDi}}, X_{\mathrm{PDe}}, X_{\mathrm{HAc}}, X_{\mathrm{EtOH}}, D, X_{\mathrm{SF}})^{\mathrm{T}}$$
$$F_{i1}(\boldsymbol{y}) = -J_{i1}(\boldsymbol{y}), i = 1, 2, 3$$
$$F_{i1}(\boldsymbol{y}) = J_{i1}(\boldsymbol{y}), i = 4, 5, 6$$
$$F_{i2}(\boldsymbol{y}) = J_{i2}(\boldsymbol{y}), i = 1, 2, \ldots, 6$$
$$H_1(\boldsymbol{y}) = \frac{X_{3-\mathrm{HPA}} + r_1}{X_{3-\mathrm{HPA}} + r_1 + r_2 X_{3-\mathrm{HPA}}} - \left(r_3 + \mu_{\mathrm{B}}\right) m_{\mathrm{R}}$$
$$H_2(\boldsymbol{y}) = r_{45} m_{\mathrm{R}} - \left(r_4 + \mu_{\mathrm{B}}\right) X_{\mathrm{R1}} - \frac{r_5 r_6 X_{3-\mathrm{HPA}}}{X_{3-\mathrm{HPA}} + r_1}$$
$$H_3(\boldsymbol{y}) = \frac{X_{3-\mathrm{HPA}} + r_1}{X_{3-\mathrm{HPA}} + r_1 + r_2 X_{3-\mathrm{HPA}}} - \left(r_7 + \mu_{\mathrm{B}}\right) m_{\mathrm{GDHt}}$$
$$H_4(\boldsymbol{y}) = \frac{X_{3-\mathrm{HPA}} + r_1}{X_{3-\mathrm{HPA}} + r_1 + r_2 X_{3-\mathrm{HPA}}} - \left(r_8 + \mu_{\mathrm{B}}\right) m_{\mathrm{PDOR}}$$
$$H_5(\boldsymbol{y}) = r_{46} m_{\mathrm{GDHt}} - \left(r_9 + \mu_{\mathrm{B}}\right) X_{[\mathrm{GDHt}]}$$
$$H_6(\boldsymbol{y}) = r_{47} m_{\mathrm{PDOR}} - \left(r_{10} + \mu_{\mathrm{B}}\right) X_{[\mathrm{PDOR}]}$$
$$H_7(\boldsymbol{y}) = \left(\mu_{\mathrm{B}} - D\right) X_{\mathrm{B}}$$
$$H_8(\boldsymbol{y}) = D\left(X_{\mathrm{SF}} - X_{\mathrm{Se}}\right) - q_{\mathrm{S}} X_{\mathrm{B}}$$
$$H_9(\boldsymbol{y}) = \frac{1}{r_{11}}\left[\frac{r_{12} X_{\mathrm{Se}}}{X_{\mathrm{Se}} + r_{13}} + \frac{1}{r_{14}}\left(X_{\mathrm{Se}} - X_{\mathrm{Si}}\right) - q_{\mathrm{S}}\right] - \mu_{\mathrm{B}} X_{\mathrm{Si}}$$

$$H_{10}(\boldsymbol{y}) = r_{15}X_{[\mathrm{GDHt}]}\frac{X_{\mathrm{Si}}}{r_{17}\left(1+\frac{X_{3-\mathrm{HPA}}}{r_{19}}\right)+X_{\mathrm{Si}}} - r_{16}X_{[\mathrm{PDOR}]}\frac{X_{3-\mathrm{HPA}}}{r_{18}+X_{3-\mathrm{HPA}}\left(1+\frac{X_{3-\mathrm{HPA}}}{r_{20}}\right)} - \mu_{\mathrm{B}}X_{3-\mathrm{HPA}} - r_{21}r_5\frac{r_6X_{3-\mathrm{HPA}}}{X_{3-\mathrm{HPA}}+r_1}$$

$$H_{11}(\boldsymbol{y}) = r_{16}X_{[\mathrm{PDOR}]}\frac{X_{3-\mathrm{HPA}}}{r_{18}+X_{3-\mathrm{HPA}}\left(1+\frac{X_{3-\mathrm{HPA}}}{r_{20}}\right)} - r_{22}\left(X_{\mathrm{PDi}}-X_{\mathrm{PDe}}\right) - \mu_{\mathrm{B}}X_{\mathrm{PDi}}$$

$$H_{12}(\boldsymbol{y}) = q_{\mathrm{PDe}}X_{\mathrm{B}} - DX_{\mathrm{PDe}}$$

$$H_{13}(\boldsymbol{y}) = q_{\mathrm{HAc}}X_{\mathrm{B}} - DX_{\mathrm{HAc}}$$

$$H_{14}(\boldsymbol{y}) = q_{\mathrm{EtOH}}X_{\mathrm{B}} - DX_{\mathrm{EtOH}}$$

$$G_1(\boldsymbol{y}) = m_{\mathrm{R}} - 10$$

$$G_2(\boldsymbol{y}) = -m_{\mathrm{R}}$$

$$G_3(\boldsymbol{y}) = X_{\mathrm{R1}} - 10$$

$$G_4(\boldsymbol{y}) = -X_{\mathrm{R1}}$$

$$G_5(\boldsymbol{y}) = m_{\mathrm{GDHt}} - 10$$

$$G_6(\boldsymbol{y}) = -m_{\mathrm{GDHt}}$$

$$G_7(\boldsymbol{y}) = m_{\mathrm{PDOR}} - 10$$

$$G_8(\boldsymbol{y}) = -m_{\mathrm{PDOR}}$$

$$G_9(\boldsymbol{y}) = X_{[\mathrm{GDHt}]} - 10$$

$$G_{10}(\boldsymbol{y}) = -X_{[\mathrm{GDHt}]}$$

$$G_{11}(\boldsymbol{y}) = X_{[\mathrm{PDOR}]} - 10$$

$$G_{12}(\boldsymbol{y}) = -X_{[\mathrm{PDOR}]}$$

$$G_{13}(\boldsymbol{y}) = X_{\mathrm{B}} - 5$$

$$G_{14}(\boldsymbol{y}) = 0.05 - X_{\mathrm{B}}$$

$$G_{15}(\boldsymbol{y}) = X_{\mathrm{Se}} - 2039$$

$$G_{16}(\boldsymbol{y}) = -X_{\mathrm{Se}}$$

$$G_{17}(\boldsymbol{y}) = X_{\mathrm{Si}} - 2039$$

$$G_{18}(\boldsymbol{y}) = -X_{\mathrm{Si}}$$

$$G_{19}(\boldsymbol{y}) = X_{3-\mathrm{HPA}} - 1000$$

$$G_{20}(\boldsymbol{y}) = -X_{3-\mathrm{HPA}}$$

$$G_{21}(\boldsymbol{y}) = X_{\mathrm{PDi}} - 939.5$$

$$G_{22}(\boldsymbol{y}) = -X_{\mathrm{PDi}}$$

$$G_{23}(\boldsymbol{y}) = X_{\mathrm{PDe}} - 939.5$$

$$G_{24}(\boldsymbol{y}) = -X_{\mathrm{PDe}}$$

$$G_{25}(\boldsymbol{y}) = X_{\mathrm{HAc}} - 1026$$

$$G_{26}(\boldsymbol{y}) = -X_{\mathrm{HAc}}$$

$$G_{27}(\boldsymbol{y}) = X_{\mathrm{EtOH}} - 360.9$$

$$G_{28}(\boldsymbol{y}) = -X_{\mathrm{EtOH}}$$

$$G_{29}(\boldsymbol{y}) = D - 0.5$$

$$G_{30}(\boldsymbol{y}) = 0.05 - D$$

$$G_{31}(\boldsymbol{y}) = X_{\mathrm{SF}} - 2039$$

$$G_{32}(\boldsymbol{y}) = -X_{\mathrm{SF}}$$

The objective functions $F_{i1}(\boldsymbol{y})$ and $F_{i2}(\boldsymbol{y})$ in Eqs. (65)–(68) have different physical meanings or order of magnitudes. Thus, the normalization for each of objective functions is carried out as follows:

$$\overline{F}_{i1}(\boldsymbol{y}) = \frac{F^*_{i1}(\boldsymbol{y}^*_{i1}) - F_{i1}(\boldsymbol{y})}{F^*_{i1}(\boldsymbol{y}^*_{i1}) - F^{\min}_{i1}}, i = 1, 2, \ldots, 6$$

$$\overline{F}_{i2}(\boldsymbol{y}) = \frac{F^*_{i2}(\boldsymbol{y}^*_{i2}) - F_{i2}(\boldsymbol{y})}{F^*_{i2}(\boldsymbol{y}^*_{i2}) - F^{\min}_{i2}}, i = 1, 2, \ldots, 6$$

where $\boldsymbol{y}^*_{ih}$ and $F^*_{ih}(\boldsymbol{y}^*_{ih})$ ($i = 1, 2, \ldots, 6$, $h = 1, 2$) are respectively the optimal solutions and corresponding objective values of single-objective problems

$$\max \quad F_{ih}(\boldsymbol{y})$$

subject to satisfying:

$$H_j(\boldsymbol{y}) = 0, j = 1, 2, \dots, 14$$
$$G_k(\boldsymbol{y}) \le 0, k = 1, 2, \dots, 32$$

And $F_{ih}^{\min}$ have the following expressions:

$$F_{i1}^{\min} = \min\{F_{i1}(\boldsymbol{y}_{i1}^*), F_{i1}(\boldsymbol{y}_{i2}^*)\}, i = 1, 2, \dots, 6$$
$$F_{i2}^{\min} = \min\{F_{i2}(\boldsymbol{y}_{i1}^*), F_{i2}(\boldsymbol{y}_{i2}^*)\}, i = 1, 2, \dots, 6$$

After the above normalization for each of objective functions, we can rewrite Eqs. (65)–(68) as:

$$\min \quad \overline{F}_{i1}(\boldsymbol{y}) \tag{69}$$
$$\min \quad \overline{F}_{i2}(\boldsymbol{y}) \tag{70}$$

subject to satisfying:

$$H_j(\boldsymbol{y}) = 0, j = 1, 2, \dots, 14 \tag{71}$$
$$G_k(\boldsymbol{y}) \le 0, k = 1, 2, \dots, 32 \tag{72}$$

Next we propose a hybrid optimization method within the NBI framework to address the proposed multi-objective models A–F. We first introduce the NBI approach. This method generates the Pareto solutions of a multi-objective problem based on the Das parameterization of the Pareto curve [39]. A set of transformed single-objective problems must be solved in the NBI method. These single-objective problems for the proposed multi-objective models A–F can be represented as:

$$\max \quad x_i \tag{73}$$

subject to satisfying:

$$\boldsymbol{\Theta}_i \boldsymbol{\alpha}_i^g + x_i \boldsymbol{n}_i = \boldsymbol{f}_i(\boldsymbol{y}) \tag{74}$$
$$H_j(\boldsymbol{y}) = 0, j = 1, 2, \dots, 14 \tag{75}$$
$$G_k(\boldsymbol{y}) \le 0, k = 1, 2, \dots, 32 \tag{76}$$

In Eqs. (73)–(76), $x_i (i = 1, 2, \dots, 6)$ is the auxiliary variable of the ith multi-objective model; $\boldsymbol{\Theta}_i$ denotes the pay-off matrix and is defined as Eq. (77); $\boldsymbol{\alpha}_i^g$ denotes the non-negative weight vector $(\alpha_{i1}^g, \alpha_{i2}^g)^{\mathrm{T}} (\alpha_{i1}^g + \alpha_{i2}^g = 1)$; $\boldsymbol{n}_i$ denotes the quasi-normal vector $\boldsymbol{\Theta}_i \cdot (-1, -1)^{\mathrm{T}}$; and $\boldsymbol{f}_i(\boldsymbol{y}) = (\overline{F}_{i1}(\boldsymbol{y}), \overline{F}_{i2}(\boldsymbol{y}))^{\mathrm{T}}$.

$$\boldsymbol{\Theta}_i = \begin{bmatrix} \overline{F}_{i1}(\boldsymbol{y}_{i1}^*) & \overline{F}_{i1}(\boldsymbol{y}_{i2}^*) \\ \overline{F}_{i2}(\boldsymbol{y}_{i1}^*) & \overline{F}_{i2}(\boldsymbol{y}_{i2}^*) \end{bmatrix} \tag{77}$$

Given a weight vector $\boldsymbol{\alpha}_i^g$, the solution to Eqs. (73)–(76) yields a Pareto optimal solution of the ith proposed multi-objective model. As we systematically modify the weight vector $\boldsymbol{\alpha}_i^g$ ($g = 1, 2, \dots, l_i$, l_i is the number of Pareto optimal solutions to the ith multi-objective model) and solve Eqs. (73)–(76), we can obtain the approximated Pareto sets and Pareto fronts of the proposed multi-objective models A–F.

To efficiently solve the single-objective problem in Eqs. (73)–(76), we present a hybrid optimization method that couples a deterministic interior point method and a stochastic genetic algorithm. In this method, the stochastic genetic algorithm is first used to solve the single-objective problem in Eqs. (73)–(76). Set the optimal solution of stochastic genetic algorithm to be the initial point of interior point method. Then we can obtain the final optimal solution of optimization problem in Eqs. (73)–(76) by using the deterministic interior point method.

Based on the similar thought of [40–42], the single-objective problem in Eqs. (73)–(76) is rewritten as the following barrier problem in the applications of deterministic interior point method:

$$\min \quad f_\lambda(\boldsymbol{y}, x_i) = -x_i - \lambda \sum_{k=1}^{32} \ln(v_k) \tag{78}$$

subject to satisfying:

$$\boldsymbol{\Theta}_i \boldsymbol{\alpha}_i^g + x_i \boldsymbol{n}_i = \boldsymbol{f}_i(\boldsymbol{y}) \tag{79}$$
$$\boldsymbol{H}(\boldsymbol{y}) = 0 \tag{80}$$
$$\boldsymbol{G}(\boldsymbol{y}) + \boldsymbol{v} = 0 \tag{81}$$

where $\boldsymbol{v} \in \mathrm{R}^{32}$ $(\boldsymbol{v} > 0)$ denote the vector of slack variables v_k $(k = 1, 2, \dots, 32)$; $\lambda > 0$ denote the barrier parameter; $\boldsymbol{H} = (H_1, H_2, \dots, H_{14})^{\mathrm{T}}$; and $\boldsymbol{G} = (G_1, G_2, \dots, G_{32})^{\mathrm{T}}$. We can see that as $\lambda \to 0$, the minimum of new objective function $f_\lambda(\boldsymbol{y}, x_i)$ will approach the minimum of $-x_i$.

Now we present the following hybrid optimization algorithm to solve the ith $(i = 1, 2, \dots, 6)$ transformed single-objective problem in Eqs. (73)–(76):

Step 0. Given the number l_i of Pareto optimal solutions, solution accuracy $\varepsilon_1 > 0$ and $\varepsilon_2 > 0$, initial barrier parameter $\lambda^1 > 0$, initial slack variable $\boldsymbol{v}^1 > 0$, and initial penalty factor $\theta^1 > 0$. Set initial counters: $s = 1$, $p = 1$, $q = 1$.

Step 1. At the s-th ($s \geq 1$) iteration, set weight vector $\boldsymbol{\alpha}_i^s$ to be

$$\boldsymbol{\alpha}_i^s = \left(\frac{s-1}{l_i - 1}, 1 - \frac{s-1}{l_i - 1}\right)^{\mathrm{T}}$$

Solve the single-objective problem in Eqs. (73)–(76) by using the stochastic genetic algorithm (For example, using the MATLAB™ software). Let $((\boldsymbol{y}_i^s)^{\mathrm{T}}, x_i^s)^{\mathrm{T}}$ be the optimal solution.

Step 2. Run the interior point algorithm by implementing Steps 2.1–2.7.

Step 2.1. Set $((\boldsymbol{y}_i^s)^{\mathrm{T}}, x_i^s, (\boldsymbol{v}^1)^{\mathrm{T}})^{\mathrm{T}}$ to be the initial point.

Step 2.2. If $p < 3$, then go to Step 2.3. Else go to Step 2.4.

Step 2.3. Solve the unconstrained problem in Eq. (82) by the direct factorization step of Waltz et al. [40]. Let $((\hat{\boldsymbol{y}}_i^p)^{\mathrm{T}}, \hat{x}_i^p, (\hat{\boldsymbol{v}}^p)^{\mathrm{T}})^{\mathrm{T}}$ be the optimal solution. Go to Step 2.5.

$$\min \quad \phi_p(\boldsymbol{y}, x_i, \boldsymbol{v}, \theta^p) = f_{\lambda^q}(\boldsymbol{y}, x_i, \boldsymbol{v}) + \theta^p \pi(\boldsymbol{y}, x_i, \boldsymbol{v}) = f_{\lambda^q}(\boldsymbol{y}, x_i, \boldsymbol{v}) + \theta^p \left\| \begin{matrix} \boldsymbol{\Theta}_i \boldsymbol{\alpha}_i^{g_i} + x_i \boldsymbol{n}_i - \boldsymbol{f}_i(\boldsymbol{y}) \\ \boldsymbol{H}(\boldsymbol{y}) \\ \boldsymbol{G}(\boldsymbol{y}) + \boldsymbol{v} \end{matrix} \right\| \tag{82}$$

Step 2.4. If $\phi_{p-1}(\hat{\boldsymbol{y}}_i^{p-1}, \hat{x}_i^{p-1}, \hat{\boldsymbol{v}}^{p-1}, \theta^{p-1}) > \phi_{p-2}(\hat{\boldsymbol{y}}_i^{p-2}, \hat{x}_i^{p-2}, \hat{\boldsymbol{v}}^{p-2}, \theta^{p-2})(p \geq 3)$, then go to Step 2.7. Else go to step 2.3.

Step 2.5. If $\pi(\hat{\boldsymbol{y}}_i^p, \hat{x}_i^p, \hat{\boldsymbol{v}}^p) < \varepsilon_1$, then go to Step 3. Else go to Step 2.6.

Step 2.6. Update $\theta^{p+1}(\theta^{p+1} > \theta^p)$. Set $p = p + 1$ and go to Step 2.2.

Step 2.7. Use the conjugate gradient step of Waltz et al. [40] to solve the problem in Eq. (82) by implementing Steps 2.7.1–2.7.2.

Step 2.7.1. Use the conjugate gradient step of Waltz et al. [40] to solve the problem in Eq. (82). Let $((\hat{\boldsymbol{y}}_i^p)^{\mathrm{T}}, \hat{x}_i^p, (\hat{\boldsymbol{v}}^p)^{\mathrm{T}})^{\mathrm{T}}$ be the optimal solution.

Step 2.7.2. If $\pi(\hat{\boldsymbol{y}}_i^p, \hat{x}_i^p, \hat{\boldsymbol{v}}^p) < \varepsilon_1$, then go to Step 3. Else reset $\theta^{p+1}(\theta^{p+1} > \theta^p)$. Set $p = p + 1$ and go to Step 2.7.1.

Step 3. If $\lambda^q < \varepsilon_2$, then record the corresponding optimum of variable $\boldsymbol{y}$ as $\overline{\boldsymbol{y}}_i^s$, and go to Step 4. Else reset λ^{q+1} so that $0 < \lambda^{q+1} < \lambda^q$. Set $q = q + 1$ and go to Step 2.2.

Step 4. If $s = l_i$, then go to Step 5. Else set $s = s + 1$ and continue from Step 1.

Step 5. Delete the non-Pareto solutions by Steps 5.1–5.8.

Step 5.1 Set initial counters: $a = 0$, $b = 0$, $c = 1$.

Step 5.2 Set $a = a + 1$; $b = 0$.

Step 5.3 Set $b = b + 1$.

Step 5.4 If $a = b$, then go to Step 5.3. Else go to Step 5.5.

Step 5.5 If $(F_{i1}(\overline{\boldsymbol{y}}_i^a), F_{i2}(\overline{\boldsymbol{y}}_i^a)) \neq (F_{i1}(\overline{\boldsymbol{y}}_i^b), F_{i2}(\overline{\boldsymbol{y}}_i^b))$ and $F_{ij}(\overline{\boldsymbol{y}}_i^a) - F_{ij}(\overline{\boldsymbol{y}}_i^b) \geq 0$, $\forall j \in \{1, 2\}$, then $\overline{\boldsymbol{y}}_i^a$ is not a Pareto optimal solution and is removed. Go to Step 5.8. Else go to Step 5.6.

Step 5.6 If $b = l_i$, then $\overline{\boldsymbol{y}}_i^a$ is a Pareto optimal solution and go to Step 5.7. Else go to Step 5.3.

Step 5.7 Set $c = c + 1$ and go to Step 5.8.

Step 5.8 If $a \neq l_i$, then go to Step 5.2. Else stop.

Remark 1. Note that in some cases the NBI framework can generate non-Pareto solutions [39,43,44]. Therefore, a simple filter strategy is given in Step 5 of the algorithm to remove those non-Pareto points generated by the NBI framework.

5. Computation results

The proposed multi-objective steady-state models A–F in Section 3 are solved by using the proposed hybrid optimization algorithm in Section 4, respectively. In the proposed hybrid optimization algorithm, the parameters λ^1, θ^1, ε_1 and ε_2 are set to be 0.1, 1, 10^{-6} and 10^{-6}, respectively. The number l_i of Pareto optimal solutions to the multi-objective models A–F is set to be 101. In Step 1 of the proposed hybrid optimization algorithm, we used the stochastic genetic algorithm of MATLAB™ software.

The Pareto fronts of multi-objective models A–F using the proposed hybrid method are presented in Figs. 1–6. Tables 1–6 present the five Pareto optimal solutions of multi-objective models A–F attained by the proposed hybrid optimization method.

Next we show a brief discussion to the Pareto fronts of Figs. 1–6 generated by the proposed hybrid optimization algorithm. From Fig. 1, we can observe that the intracellular 3-HPA concentration $X_{3-\mathrm{HPA}}$ and the production rate DX_{PDe} of extracellular 1,3-PDO simultaneously increase or decrease in values. That is to say, we cannot find a solution that minimizes the intracellular 3-HPA concentration $X_{3-\mathrm{HPA}}$ while maximizing the production rate DX_{PDe} of extracellular 1,3-PDO. The maximum for production rate DX_{PDe} of extracellular 1,3-PDO can be 114.3005 mmol/(L·h), which is obtained when the intracellular 3-HPA concentration $X_{3-\mathrm{HPA}}$ is 37.5360 mmol/L. Also in Fig. 1, there are three discontinuous parts in the Pareto curve. In Fig. 2, we can see that the conversion rate $X_{\mathrm{PDe}}/(X_{\mathrm{SF}} - X_{\mathrm{Se}})$ of extracellular glycerol to extracellular 1,3-PDO increases rapidly from 0.5009 to 0.6231, when the intracellular 3-HPA concentration $X_{3-\mathrm{HPA}}$ increases from 15.3080 to 21.9629 mmol/L. Also in Fig. 2, we can observe that the conversion rate $X_{\mathrm{PDe}}/(X_{\mathrm{SF}} - X_{\mathrm{Se}})$ of extracellular glycerol to extracellular 1,3-PDO increases slowly from 0.6231 to 0.7374, when the intracellular 3-HPA concentration $X_{3-\mathrm{HPA}}$ increases from 21.9629 to 139.8691 mmol/L. From Fig. 3, we can see that as the conversion rate $(X_{\mathrm{SF}} - X_{\mathrm{Se}})/X_{\mathrm{SF}}$ of extracellular glycerol increases from 0.9973 to 0.9999, the intracellular 3-HPA concentration increases from 9.3689 to 20.4801 mmol/L. In Fig. 3, we can also observe that the conversion rate $(X_{\mathrm{SF}} - X_{\mathrm{Se}})/X_{\mathrm{SF}}$ of extracellular

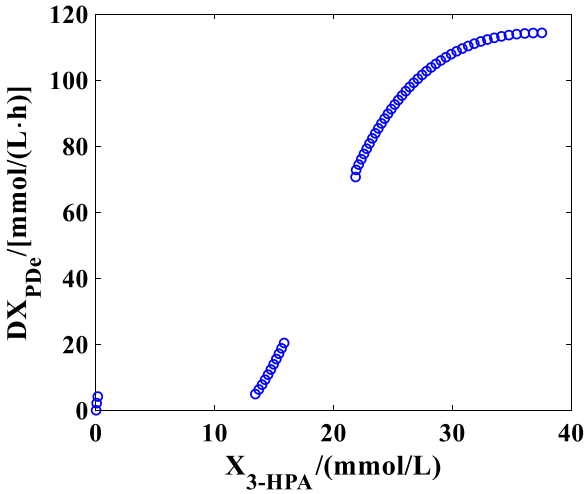

Fig. 1. Pareto front of multi-objective model A.

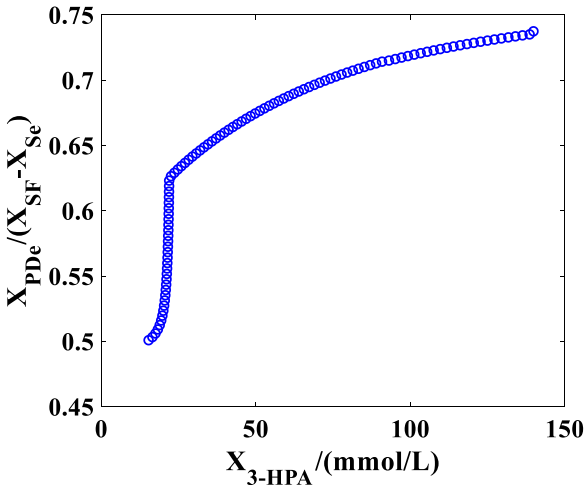

Fig. 2. Pareto front of multi-objective model B.

glycerol almost achieves its desired maximum 1.0. From Fig. 4, we can see that the production rate DX_{PDe} of extracellular 1,3-PDO and the conversion rate $X_{\mathrm{PDe}}/(X_{\mathrm{SF}} - X_{\mathrm{Se}})$ of extracellular glycerol to extracellular 1,3-PDO cannot reach their respective maximum values at the same time. This is consistent with the existing experimental results of Biebl et al. [1]. When the value of the conversion rate $X_{\mathrm{PDe}}/(X_{\mathrm{SF}} - X_{\mathrm{Se}})$ of extracellular glycerol to extracellular 1,3-PDO is 0.6324, the maximum for production rate DX_{PDe} of extracellular 1,3-PDO can be 114.3005 mmol/(L·h). From Fig. 5, we can observe that as the conversion rate $(X_{\mathrm{SF}} - X_{\mathrm{Se}})/X_{\mathrm{SF}}$ of extracellular glycerol increases from 0.8658 to 0.9999, the production rate DX_{PDe} of extracellular 1,3-PDO decreases from 114.3005 to 1.4858 mmol/(L·h). Also in Fig. 5, we can obtain a larger value between 100.0986 and 114.3005 mmol/(L·h) for the production rate DX_{PDe} of extracellular 1,3-PDO if the conversion rate $(X_{\mathrm{SF}} - X_{\mathrm{Se}})/X_{\mathrm{SF}}$ of extracellular glycerol reaches any value in the interval [0.8658, 0.9883]. From Fig. 6, we can observe that the conversion rate $X_{\mathrm{PDe}}/(X_{\mathrm{SF}} - X_{\mathrm{Se}})$ of extracellular glycerol to extracellular 1,3-PDO and conversion rate $(X_{\mathrm{SF}} - X_{\mathrm{Se}})/X_{\mathrm{SF}}$ of extracellular glycerol cannot reach their respective maximum values 0.7374 and 0.9999 at the same time. As the conversion rate $(X_{\mathrm{SF}} - X_{\mathrm{Se}})/X_{\mathrm{SF}}$ of extracellular glycerol decreases from 0.9999 to 0.5062, the conversion rate $X_{\mathrm{PDe}}/(X_{\mathrm{SF}} - X_{\mathrm{Se}})$ of extracellular glycerol to extracellular 1,3-PDO increases from 0.0921 to 0.7353. Additionally, as the conversion rate $(X_{\mathrm{SF}} - X_{\mathrm{Se}})/X_{\mathrm{SF}}$ of extracellular glycerol decreases from 0.0960 to 0.0191, the conversion rate $X_{\mathrm{PDe}}/(X_{\mathrm{SF}} - X_{\mathrm{Se}})$ of extracellular glycerol to extracellular 1,3-PDO increases from 0.7364 to 0.7374.

Figs. 7–9 show the comparison of Pareto fronts obtained by the proposed hybrid optimization algorithm and the reported method [25] for multi-objective models D–F. As can be seen in Figs. 7–9, the proposed hybrid optimization algorithm can find the better Pareto fronts of multi-objective models D–F than the existing method. This concludes that the proposed hybrid algorithm is a good choice in handling the multi-objective steady-state optimization of the complex bioprocess in glycerol metabolism.

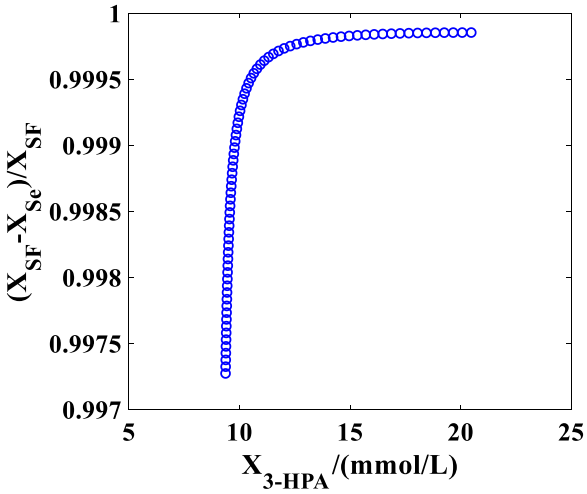

Fig. 3. Pareto front of multi-objective model C.

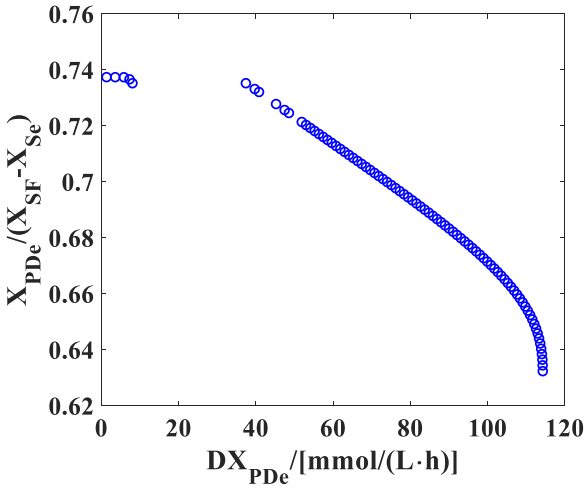

Fig. 4. Pareto front of multi-objective model D.

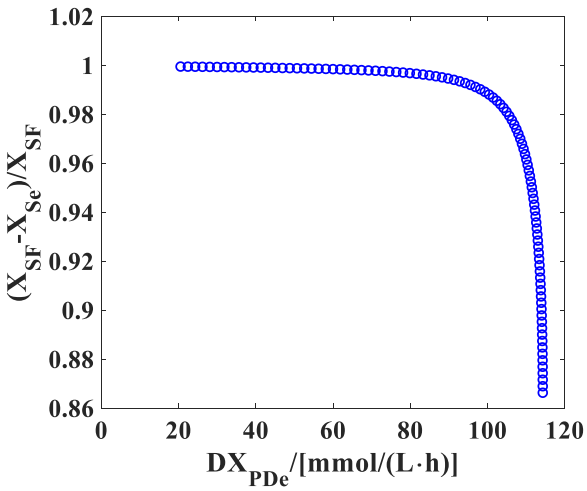

Fig. 5. Pareto front of multi-objective model E.

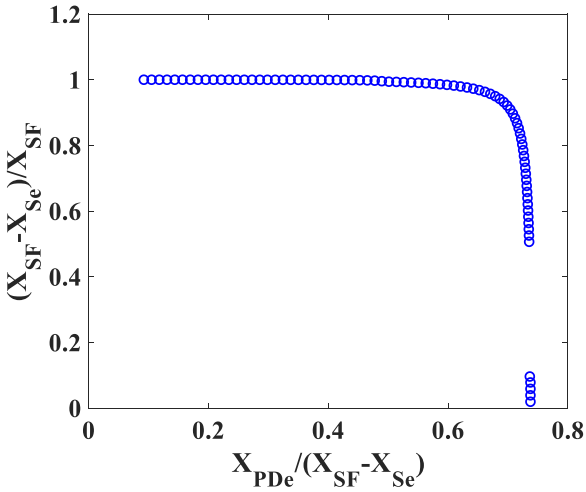

Fig. 6. Pareto front of multi-objective model F.

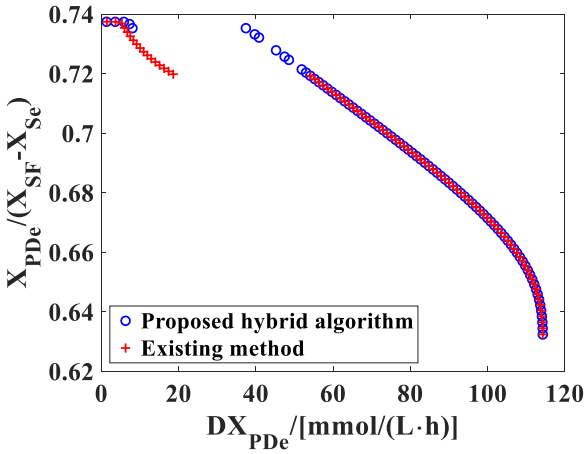

Fig. 7. Pareto front of multi-objective model D.

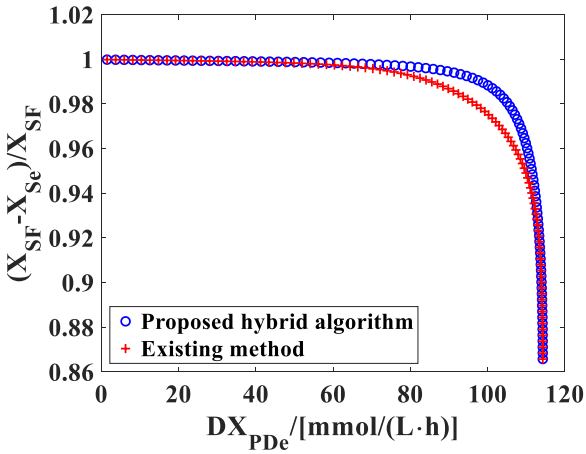

Fig. 8. Pareto front of multi-objective model E.

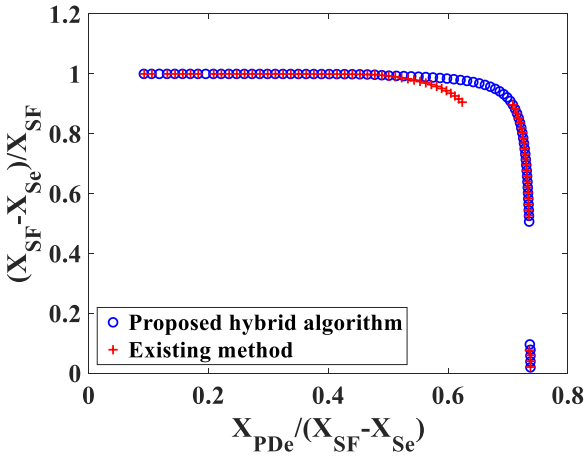

Fig. 9. Pareto front of multi-objective model F.

Table 1
Five pareto optimal solutions of multi-objective model A.

Variable	Solution 1	Solution 2	Solution 3	Solution 4	Solution 5	Unit
m_R	6.2631	1.6943	1.2635	1.4439	1.6678	h
X_{R1}	1.1102×10^{-5}	3.0032×10^{-6}	2.2395×10^{-6}	2.5592×10^{-6}	2.9561×10^{-6}	mmol/L
m_{GDHt}	0.6262	0.4284	0.3747	0.3714	0.3599	h
m_{PDOR}	0.0870	0.0697	0.0642	0.0608	0.0563	h
$X_{[GDHt]}$	0.5274	0.3518	0.3050	0.3047	0.2976	mmol/L
$X_{[PDOR]}$	0.1035	0.0816	0.0746	0.0710	0.0661	mmol/L
X_B	0.0500	0.6590	1.4583	2.2900	2.8857	g/L
X_{Se}	0.0229	0.4784	38.9823	70.2555	98.0997	mmol/L
X_{Si}	0.0215	0.4718	38.9757	70.2508	98.0973	mmol/L
X_{3-HPA}	0.0290	15.1914	21.9049	28.2169	37.5360	mmol/L
X_{PDi}	0.7619	40.6882	143.8898	263.6917	395.6073	mmol/L
X_{PDe}	0.7337	41.2939	146.7274	267.7582	400.1021	mmol/L
X_{HAc}	0.6850	20.1516	50.6491	85.2706	116.5758	mmol/L
X_{EtOH}	4.2737	17.7996	18.8226	29.5624	42.3266	mmol/L
D	0.0500	0.3765	0.4961	0.3878	0.2857	h^{-1}
X_{SF}	8.3776	86.7069	288.2450	507.6628	730.7995	mmol/L
J_{11} (X_{3-HPA})	0.0290	15.1914	21.9049	28.2169	37.5360	mmol/L
J_{12} (DX_{PDe})	0.0367	15.5473	72.7943	103.8411	114.3005	mmol/(L·h)

Table 2
Five pareto optimal solutions of multi-objective model B.

Variable	Solution 1	Solution 2	Solution 3	Solution 4	Solution 5	Unit
m_R	1.3493	1.2636	1.2547	2.2079	2.2870	h
X_{R1}	2.3916×10^{-6}	2.2397×10^{-6}	2.2239×10^{-6}	3.9129×10^{-6}	4.0526×10^{-6}	mmol/L
m_{GDHt}	0.4020	0.3764	0.3738	0.2916	0.2287	h
m_{PDOR}	0.0689	0.0645	0.0641	0.0418	0.0318	h
$X_{[GDHt]}$	0.3271	0.3063	0.3041	0.2445	0.1926	mmol/L
$X_{[PDOR]}$	0.0801	0.0750	0.0745	0.0496	0.0378	mmol/L
X_B	0.3818	1.5140	0.6006	0.0507	0.0506	g/L
X_{Se}	1.0661	14.4452	345.6739	1700.2560	1880.6723	mmol/L
X_{Si}	1.0582	14.4395	345.6620	1700.2527	1880.6730	mmol/L
X_{3-HPA}	15.3080	21.2905	21.9629	81.3950	138.6283	mmol/L
X_{PDi}	24.6134	130.5827	66.6687	14.3832	27.3147	mmol/L
X_{PDe}	25.0994	133.1777	67.9925	14.4452	27.3689	mmol/L
X_{HAc}	11.9406	49.6380	24.2216	3.7697	6.2423	mmol/L
X_{EtOH}	7.2034	22.3437	2.5472	0.4513	1.3403	mmol/L
D	0.5000	0.5000	0.5000	0.1090	0.0500	h^{-1}
X_{SF}	51.1745	254.0553	454.7917	1720.6792	1917.7867	mmol/L
J_{21} (X_{3-HPA})	15.3080	21.2905	21.9629	81.3950	139.8691	mmol/L
J_{22} $[X_{PDe}/(X_{SF}-X_{Se})]$	0.5009	0.5558	0.6231	0.7073	0.7374	–

Table 3
Five pareto optimal solutions of multi-objective model C.

Variable	Solution 1	Solution 2	Solution 3	Solution 4	Solution 5	Unit
m_R	5.5305	5.5209	5.4818	5.1523	4.8680	h
X_{R1}	9.8037×10^{-6}	9.7866×10^{-6}	9.7173×10^{-6}	9.1331×10^{-6}	8.6290×10^{-6}	mmol/L
m_{GDHt}	0.5530	0.5520	0.5481	0.5152	0.4867	h
m_{PDOR}	0.0769	0.0767	0.0762	0.0716	0.0677	h
$X_{[GDHt]}$	0.4657	0.4649	0.4616	0.4339	0.4099	mmol/L
$X_{[PDOR]}$	0.0914	0.0912	0.0905	0.0851	0.0804	mmol/L
X_B	0.0502	0.0808	0.2067	1.2192	1.9172	g/L
X_{Se}	0.0229	0.0231	0.0240	0.0341	0.0471	mmol/L
X_{Si}	0.0215	0.0217	0.0226	0.0328	0.0458	mmol/L
X_{3-HPA}	9.3689	9.5095	10.0862	15.3293	20.4801	mmol/L
X_{PDi}	0.7412	1.1910	3.0413	18.3258	29.6603	mmol/L
X_{PDe}	0.7361	1.1869	3.0413	18.3585	29.7166	mmol/L
X_{HAc}	0.6872	1.1076	2.8325	16.7219	26.3295	mmol/L
X_{EtOH}	4.2874	6.9092	17.6625	103.7974	162.4974	mmol/L
D	0.0500	0.0500	0.0500	0.0500	0.0500	h^{-1}
X_{SF}	8.4043	13.5318	34.5771	204.4322	322.7087	mmol/L
J_{31} (X_{3-HPA})	9.3689	9.5095	10.0862	15.3293	20.4801	mmol/L
J_{32} $[(X_{SF}-X_{Se})/C_{SF}]$	0.9973	0.9983	0.9993	0.9998	0.9999	–

Table 4
Five pareto optimal solutions of multi-objective model D.

Variable	Solution 1	Solution 2	Solution 3	Solution 4	Solution 5	Unit
m_R	2.2869	2.2922	2.2158	1.9389	1.6678	h
X_{R1}	4.0526×10^{-6}	4.06203×10^{-6}	3.92694×10^{-6}	3.4364×10^{-6}	2.9561×10^{-6}	mmol/L
m_{GDHt}	0.2287	0.2320	0.2899	0.3350	0.3599	h
m_{PDOR}	0.0318	0.0323	0.0415	0.0500	0.0563	h
$X_{[GDHt]}$	0.1926	0.1954	0.2431	0.2791	0.2976	mmol/L
$X_{[PDOR]}$	0.0378	0.0384	0.0527	0.0591	0.0661	mmol/L
X_B	0.0500	1.4737	2.5577	2.8402	2.8857	g/L
X_{Se}	1880.7658	847.9711	451.6410	305.9519	98.0985	mmol/L
X_{Si}	1880.7665	847.9732	451.6410	305.9505	98.0960	mmol/L
X_{3-HPA}	139.8822	135.8912	70.9039	53.0644	37.5360	mmol/L
X_{PDi}	27.0129	759.6273	618.4141	525.5838	395.6074	mmol/L
X_{PDe}	27.0665	761.2041	621.6429	529.5649	400.1022	mmol/L
X_{HAc}	6.1732	168.5584	159.6900	146.3295	116.5758	mmol/L
X_{EtOH}	1.3256	57.3735	42.7651	37.9821	42.3269	mmol/L
D	0.0500	0.0522	0.1313	0.1904	0.2857	h^{-1}
X_{SF}	1917.4698	1886.1738	1349.5755	1096.0055	730.7987	mmol/L
J_{41} (DX_{PDe})	1.3533	37.4264	70.7221	100.8524	114.3005	mmol/(L·h)
J_{42} $[X_{PDe}/(X_{SF}-X_{Se})]$	0.7374	0.7353	0.7032	0.6703	0.6324	–

Table 5
Five pareto optimal solutions of multi-objective model E.

Variable	Solution 1	Solution 2	Solution 3	Solution 4	Solution 5	Unit
m_R	6.2579	2.3137	1.7599	1.6848	1.6688	h
X_{R1}	1.1093×10^{-5}	4.1010×10^{-6}	3.1193×10^{-6}	2.9862×10^{-6}	2.9578×10^{-6}	mmol/L
m_{GDHt}	0.6257	0.3885	0.3611	0.3597	0.3598	h
m_{PDOR}	0.0870	0.0577	0.0558	0.0561	0.0562	h
$X_{[GDHt]}$	0.5270	0.3239	0.2992	0.2976	0.2976	mmol/L
$X_{[PDOR]}$	0.1034	0.0682	0.0656	0.0659	0.0661	mmol/L
X_B	1.9158	4.4731	4.0936	3.1504	2.8872	g/L
X_{Se}	0.0470	0.7465	8.5606	41.3379	98.2132	mmol/L
X_{Si}	0.0458	0.7453	8.5607	41.3365	98.2108	mmol/L
X_{3-HPA}	0.0825	35.5896	38.7047	37.9138	37.5795	mmol/L
X_{PDi}	29.7116	264.4576	379.2532	395.8866	396.1265	mmol/L
X_{PDe}	29.7103	266.3529	383.1932	400.3015	400.6213	mmol/L
X_{HAc}	26.3162	125.0859	128.3417	114.2691	116.6831	mmol/L
X_{EtOH}	162.3677	148.5893	79.2267	56.5605	42.3783	mmol/L
D	0.0500	0.1803	0.2612	0.2804	0.2853	h^{-1}
X_{SF}	322.4474	644.3106	734.0439	701.7973	731.6514	mmol/L
J_{51} (DX_{PDe})	1.4858	48.0158	100.0986	112.2478	114.3005	mmol/(L·h)
J_{52} $[(X_{SF}-X_{Se})/C_{SF}]$	0.9999	0.9988	0.9883	0.9411	0.8658	–

Table 6
Five pareto optimal solutions of multi-objective model F.

Variable	Solution 1	Solution 2	Solution 3	Solution 4	Solution 5	Unit
m_R	6.2583	2.6114	2.2926	2.2873	2.2869	h
X_{R1}	1.1094×10^{-5}	4.6289×10^{-6}	4.0627×10^{-6}	4.0532×10^{-6}	4.0526×10^{-6}	mmol/L
m_{GDHt}	0.6257	0.4387	0.2292	0.2287	0.2287	h
m_{PDOR}	0.0870	0.0652	0.0319	0.0318	0.0318	h
$X_{[GDHt]}$	0.5270	0.3657	0.1931	0.1926	0.1926	mmol/L
$X_{[PDOR]}$	0.1034	0.0770	0.0379	0.0378	0.0378	mmol/L
X_B	1.9172	2.4785	1.6888	1.3921	0.0500	g/L
X_{Se}	0.0471	0.2721	115.5351	992.6461	1880. 7684	mmol/L
X_{Si}	0.0458	0.2692	115.5387	992.6481	1880. 7691	mmol/L
X_{3-HPA}	0.0826	22.8951	139.2545	139.8479	139.8850	mmol/L
X_{PDi}	29.7180	136.1532	813.7154	746.7629	27.0140	mmol/L
X_{PDe}	29.7166	137.1284	815.3337	748.2481	27.0676	mmol/L
X_{HAc}	26.3296	68.8925	134.3907	166.1353	6.1734	mmol/L
X_{EtOH}	162.4981	113.3085	111.7091	53.1831	1.3257	mmol/L
D	0.0500	0.1805	0.0500	0.0500	0.0500	h^{-1}
X_{SF}	322.7101	341.8741	1274. 2207	2010. 2993	1917. 4758	mmol/L
J_{61} $[X_{PDe}/(X_{SF}-X_{Se})]$	0.0921	0.4014	0.7037	0.7353	0.7374	–
J_{62} $[(X_{SF}-X_{Se})/C_{SF}]$	0.9999	0.9992	0.9093	0.5062	0.0191	–

Table A.1
The parameter values in Eqs. (1)–(19), (21), (24), and (25).

Parameter	Value	Unit	Parameter	Value	Unit
r_1	917.9722	mmol/L	r_{25}	2039	mmol/L
r_2	13.1567	–	r_{26}	939.5	mmol/L
r_3	0.1096	h^{-1}	r_{27}	1026	mmol/L
r_4	1.1305×10^{5}	h^{-1}	r_{28}	360.9	mmol/L
r_5	79.9150	h^{-1}	r_{29}	2.20	mmol/(g·h)
r_6	1.48×10^{-5}	mmol/L	r_{30}	0.0082	g/mmol
r_7	1.5462	h^{-1}	r_{31}	28.58	mmol/(g·h)
r_8	11.4334	h^{-1}	r_{32}	11.43	mmol/L
r_9	12.7837	h^{-1}	r_{33}	−2.69	mmol/(g·h)
r_{10}	20.2310	h^{-1}	r_{34}	67.69	mmol/g
r_{11}	0.151	L/g	r_{35}	26.59	mmol/(g·h)
r_{12}	54.664	mmol/(g·h)	r_{36}	15.50	mmol/L
r_{13}	1.340	mmol/L	r_{37}	−0.97	mmol/(g·h)
r_{14}	1.896×10^{-4}	g/(L·h)	r_{38}	33.07	mmol/g
r_{15}	36.3625	h^{-1}	r_{39}	5.74	mmol/(g·h)
r_{16}	42.5261	h^{-1}	r_{40}	85.71	mmol/L
r_{17}	0.53	mmol/L	r_{41}	0.025	mmol/(L·h)
r_{18}	0.14	mmol/L	r_{42}	5.18	mmol/(L·h)
r_{19}	220.319	mmol/L	r_{43}	0.06	mmol/(L·h)
r_{20}	0.418	mmol/L	r_{44}	50.45	mmol/(L·h)
r_{21}	2.0421	–	r_{45}	0.2004	mmol/(L·h^2)
r_{22}	25.137	h^{-1}	r_{46}	10.8083	h^{-1}
r_{23}	0.67	h^{-1}	r_{47}	24.1030	h^{-1}
r_{24}	0.28	mmol/L	–	–	–

6. Conclusions

In this study, we have addressed the multi-objective steady-state optimization problem of complex bioconversion process of glycerol metabolism producing 1,3-PDO. The studied bioconversion system includes the regulation for *dha* regulon, enzyme-catalytic kinetics, the inhibition of intracellular 3-HPA, and so on. Several new multi-objective steady-state models have been proposed by considering the minimization of the intracellular 3-HPA concentration X_{3-HPA}, the maximization of the production rate DX_{PDe} of extracellular 1,3-PDO, the maximization of the conversion rate $X_{PDe}/(X_{SF} - X_{Se})$ of extracellular glycerol to extracellular 1,3-PDO, and the maximization of the conversion rate $(X_{SF} - X_{Se})/X_{SF}$ of extracellular glycerol. A hybrid optimization algorithm has been proposed to solve the presented complex multi-objective models. The obtained Pareto optimal solutions have been shown and discussed. This study can provide a guide to the process optimization of glycerol metabolism producing 1,3-PDO. A comparative study shown that the presented hybrid algorithm can obtain the better Pareto fronts than the existing method.

In the future study, we will deal with the process optimization and control problem of glycerol metabolism producing 1,3-PDO by considering more complex mechanisms in the intracellular environment. Additionally, we will also develop more efficient optimization algorithms to solve the proposed multi-objective steady-state optimization problems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. The parameter values in Eqs. (1)–(19), (21), (24), and (25)

See Table A.1.

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